Probability and Statistics

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Part I
Math Fundamentals

1 Sequence

Definition 1.1: Sequence

An ordered **list** of items. The items might or might not have any relationship with each other.

Example 1.1: A Simple Sequence

The following is a sequence:

$$S = a, \text{name}, 10, ?$$

Note: the items of the sequence don't have any relationship among each other, but it is a valid sequence. The individual items of the sequence are called terms of the sequence i.e a is the first term and "?" is the fourth term.

From Ex. 1.1 it is clear that from a sequence it is possible to retrieve items based on their term number. More clearly, if the term number k is given the sequence returns the value at the term number k, which is represented as S_k. This view of the sequence justifies another definition of the sequence.

Definition 1.2: Sequence as a Function

A sequence is a function from positive integers (term numbers) to another set S. Formally,

$$Sequence : n \mapsto S, n \in Z^+$$

1.1 Geometric Sequence

Items in a sequence can have a relationship with each other. Geometric Sequence is where the items in a sequence have a special relationship with each other.

Definition 1.3: Geometric Sequence

S is a geometric sequence if each subsequent terms in the sequence can be found by multiplying the previous term with a common ratio r. Formally,

$$S_n = rS_{n-1} = ar^n$$

In Def. 1.3 a is a constant. The geometric sequence is also known as the geometric progression.

Example 1.2: Geometric Sequence Example

The following is an example of an infinite geometric sequence where $r = 2$,

$$S = 1, 2, 4, 8, 16, \ldots$$

1.2 Arithmetic Sequence

Definition 1.4: Arithmetic Sequence

S is a arithmetic sequence if each subsequent term in the sequence can be found by adding the previous term with a common number d. Formally,

$$S_n = S_{n-1} + d$$

Arithmetic sequence is also know as arithmetic progression.

Example 1.3: Arithmetic Sequence Example

The following is an example of an infinite arithmetic sequence where $d = 2$,

$$S = 1, 3, 4, 7, 8, 11, \ldots$$

1.3 Sum of Geometric Progression

Theorem 1.1: Sum of Finite Geometric Progression

If $r \neq 1$, then sum of finite geometric progression is,

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{(n+1)}}{1 - r}$$

Proof. Consider the product of $(1 - r)$ and the finite geometric series:

$$\begin{aligned}
(1 - r) \sum_{k=0}^{n} r^k &= (1 - r)(1 + r + r^2 + \cdots + r^n) \\
&= (1 + r + r^2 + \cdots + r^n) - r(1 + r + r^2 + \cdots + r^n) \\
&= (1 + r + r^2 + \cdots + r^n) - (r + r^2 + r^3 + \cdots + r^{n+1}) \\
&= 1 - r^{n+1}
\end{aligned}$$

Dividing both sides by $(1 - r)$, provided $r \neq 1$, yields:

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}$$

■

Remark

The general formula to find any term in a geometric sequence is ar^k, where k is the term number. The sum of the geometric progression,

$$\sum_{k=0}^{n} ar^k = a \sum_{k=0}^{n} r^k = a\frac{1-r^{n+1}}{1-r}$$

The theorem can be extended for the infinite geometric series as well, under certain constraint.

Corollary 1.1: Sum of infinite geometric progression

If $0 < |r| < 1$, then

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$$

Proof. It is already known that,

$$\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$$

Now, if n approaches infinity, the following can be written,

$$\lim_{n\to\infty} \sum_{k=0}^{n} r^k = \lim_{n\to\infty} \frac{1-r^{n+1}}{1-r}$$

Since the value of the common ratio r is a fraction, the value of r becomes 0 when the power n is large.

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$$

■

Similarly,

$$\sum_{k=0}^{\infty} ar^k = a\frac{1}{1-r}$$

Remark

In Corollary 1.1, if $|r| > 1$ then the series diverges (means the value does not move towards a definite limiting value). For $|r| = 1$ then the denominator becomes 0. For $|r| = 0$, the sum becomes 1.

Example 1.4: Sum of infinite geometric series

Evaluate $\sum_{k=2}^{\infty} \frac{1}{2^k}$

$$\sum_{k=2}^{\infty} \frac{1}{2^k} = \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + ... = \frac{1}{2^2}[1 + \frac{1}{2} + \frac{1}{2^2} + ...] = \frac{1}{2^2}\frac{1}{1-\frac{1}{2}} = \frac{1}{2}$$

Another corollary can be derived by taking the derivative of the corollary 1.1,

Corollary 1.2: Sum of infinite geometric progression

If $0 < |r| < 1$, then

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}$$

2 Combination

Definition 2.1: Combination

A combination is one specific selection of k items from a set of n items, where order is not important. The number of all possible combinations is expressed by,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Example 2.1: Combination Example

The number of subsets of the set 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 having 3 elements can be found using combination.

$$\binom{n}{k} = \binom{10}{3} = \frac{10!}{3!(10-3)!} = 120$$

Here {1,2,3} is one of the 120 combinations.

2.1 Pascals identity

Definition 2.2: Pascals identity

If $k \leq n$ and $k, n \in Z^+$,

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

Pascals identity constructs the pascals triangle where n is the row number and k is the column number.

Example 2.2: Pascals triangle

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

2.2 Binomial series

Definition 2.3: Binomial series

Polynomials (algebric expressions with non-negative integer exponents) of the for $(a+b)^n$ where $a, b \in R$,

$$(a+b)^n = \sum_{k=0}^{n} a^{n-k} b^k$$

Example 2.3: Binomial series example

The following expression,

$$\sum_{k=0}^{n} \binom{n}{k} p^{n-k} (1-p)^k$$

can be written as $(p+1-p)^n$ using Def. 2.3, which evaluates to 1.

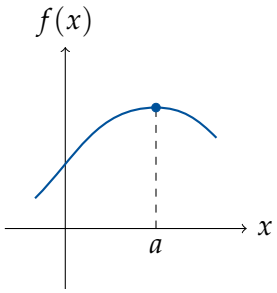
3 Approximation

3.1 Continuous Function

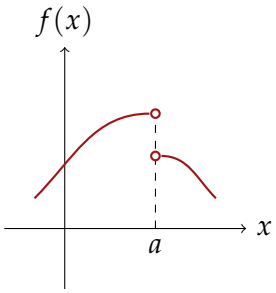
Definition 3.1: Continuous Function

All three of the following must hold for a continuous function:

1. The function has a value at every point a.
2. $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$
3. $\lim_{x \to a} f(x) = f(a)$



Continuous function



Discontinuous at a. Violates condition 1 and 2.

3.2 Taylor Approximation

Definition 3.2: Taylor Approximation

If $f : \mathbb{R} \to \mathbb{R}$ is a continuous function for which infinite derivatives exist, then the approximation of the function at a is:

$$\begin{aligned} f(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \end{aligned}$$

Remark

- The Taylor polynomials capture local features of function f at point x. As the degree of polynomial is increased, the approximation becomes better.
- With higher degree of polynomials, more local features (slope, curvature) are captured.
- If $a = 0$, the Taylor polynomial becomes a **Maclaurin Polynomial**.

Example 3.1: Taylor Approximation of Sine

Let $f(x) = \sin(x)$ at $a = 0$.

$$\begin{aligned} f(x) &\approx \sin(0) + \cos(0)x + \frac{1}{2!}(-\sin(0))x^2 + \frac{1}{3!}(-\cos(0))x^3 \\ &= 0 + x - 0 - \frac{1}{6}x^3 \\ &= x - \frac{1}{6}x^3 \end{aligned}$$

This can be expanded further:

$$f(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \frac{1}{9!}x^9 - \dots$$

Example 3.2: Taylor Approximation of Sine at $a = \frac{\pi}{4}$

Let $f(x) = \sin(x)$ at $a = \frac{\pi}{4}$.

$$\begin{aligned} f(x) &= \sin\left(\frac{\pi}{4}\right) + \cos\left(\frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right) - \frac{1}{2!}\sin\left(\frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{3!}\cos\left(\frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right)^3 \\ &= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right) - \frac{1}{2}\frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{6}\frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right)^3 \end{aligned}$$

3.3 Exponential Series

Definition 3.3: Exponential Series

For real x,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{1}{k!}x^k$$

Proof. Let $f(x) = e^x$ and $a = 0$. Using Taylor approximation:

$$\begin{aligned} f(x) &= f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 + \frac{f'''(0)}{3!}(x-0)^3 + \dots \\ &= e^0 + e^0 x + \frac{1}{2!}e^0 x^2 + \frac{1}{3!}e^0 x^3 + \dots \\ &= 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots \\ &= \sum_{k=0}^{\infty} \frac{1}{k!}x^k \end{aligned}$$

■

Example 3.3: Sum of Poisson Probabilities

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

Example 3.4: Euler's Formula via Taylor Approximation

Using Taylor approximation $x = j\theta$ for sine and cosine:

$$\begin{aligned} e^{j\theta} &= 1 + (j\theta) + \frac{1}{2!}(j\theta)^2 + \frac{1}{3!}(j\theta)^3 + \dots \\ &= 1 + j\theta - \frac{1}{2!}\theta^2 - j\frac{1}{3!}\theta^3 + \dots \\ &= \left(1 - \frac{1}{2!}\theta^2 + \dots\right) + j\left(\theta - \frac{1}{3!}\theta^3 + \dots\right) \end{aligned}$$

Since $e^{j\theta} = \cos(\theta) + j\sin(\theta)$, we can equate the real and imaginary parts:

$$\begin{aligned} \cos(\theta) &= 1 - \frac{1}{2!}\theta^2 + \dots \\ \sin(\theta) &= \theta - \frac{1}{3!}\theta^3 + \dots \end{aligned}$$

3.4 Logarithmic Approximation

Definition 3.4: Logarithmic Approximation

$$\log(1 + x) = x - \frac{x^2}{2} + \mathcal{O}(x^3)$$

Proof. Using Taylor approximation at $a = 0$ for $f(x) = \log(1 + x)$:

$$\begin{aligned} f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots \\ &= 0 + \left(\frac{1}{1+0}\right)x + \frac{\left(\frac{-1}{(1+0)^2}\right)}{2!}x^2 + \frac{\left(\frac{2}{(1+0)^3}\right)}{3!}x^3 + \dots \\ &= x - \frac{x^2}{2!} + \mathcal{O}(x^3) \end{aligned}$$

■

Example 3.5: Limit involving Logarithmic Approximation

Evaluate $\lim_{N\to\infty}\left(1+\frac{S^2}{2N}\right)^N = e^{S^2/2}$
Taking the natural logarithm of the expression inside the limit gives:

$$N\log\left(1+\frac{S^2}{2N}\right)\approx N\left(\frac{S^2}{2N}\right)=\frac{S^2}{2}$$

Exponentiating the result yields $e^{S^2/2}$.

4 Integration

4.1 Substitution

Definition 4.1: u-Substitution

A technique to evaluate integrals of the form $\int f(g(x))g'(x)dx$. The idea is to change the variable to the inner function and simplify. Let $u=g(x)$, then $du=g'(x)dx$.

$$\int f(g(x))g'(x)dx=\int f(u)du=F(u)+C=F(g(x))+C$$

Remark

- u-substitution changes the variable x to a function u.
- u-substitution is the opposite of the chain rule derivative.

Example 4.1: Simple Polynomial Substitution

Evaluate $\int (x^2+1)^4 2x\,dx$.
Let $u=x^2+1$. Then $\frac{du}{dx}=2x \implies du=2x\,dx$.

$$\begin{aligned}\int (x^2+1)^4 2x\,dx &= \int u^4\,du\\ &= \frac{u^5}{5}+C\\ &= \frac{(x^2+1)^5}{5}+C\end{aligned}$$

Example 4.2: Substitution with Algebraic Manipulation

Evaluate $\int x^2\sqrt{x-1}\,dx$.
Let $u = x-1$. Then $du = dx$ and $x = u+1$. Squaring x, we get $x^2 = (u+1)^2 = u^2+2u+1$.

$$\begin{aligned}\int x^2\sqrt{x-1}\,dx &= \int (u^2+2u+1)\sqrt{u}\,du \\ &= \int (u^2+2u+1)u^{1/2}\,du \\ &= \int \left(u^{5/2}+2u^{3/2}+u^{1/2}\right)du \\ &= \frac{2u^{7/2}}{7}+\frac{4u^{5/2}}{5}+\frac{2u^{3/2}}{3}+C \\ &= \frac{2(x-1)^{7/2}}{7}+\frac{4(x-1)^{5/2}}{5}+\frac{2(x-1)^{3/2}}{3}+C\end{aligned}$$

4.2 Integration By Parts

Definition 4.2: Integration By Parts

Simplifies integrals where two functions are multiplied together. The formula is:

$$\int u\cdot v\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$$

Alternatively, written as: $\int u\,dv = uv - \int v\,du$.

Remark

The function u is chosen using the following priority order (**LIATE**):

1. **L**ogarithmic (e.g., $\log(x)$) — *High Priority for u*
2. **I**nverse trigonometric (e.g., $\arcsin(x)$)
3. **A**lgebraic (e.g., x^3)
4. **T**rigonometric (e.g., $\sin(x)$)
5. **E**xponential (e.g., e^{3x}) — *Low Priority for u*

Example 4.3: Integration by Parts Example

Evaluate $\int xe^x\,dx$.
Let $u = x$ (Algebraic) and $dv = e^x dx$ (Exponential). Then $du = dx$ and $v = \int e^x dx = e^x$.

$$\begin{aligned}\int xe^x\,dx &= xe^x - \int e^x\,dx \\ &= xe^x - e^x + C \\ &= e^x(x-1) + C\end{aligned}$$

4.3 Even and Odd Functions

Definition 4.3: Even Functions

A function $f : \mathbb{R} \to \mathbb{R}$ is even if for all $x \in \mathbb{R}$,

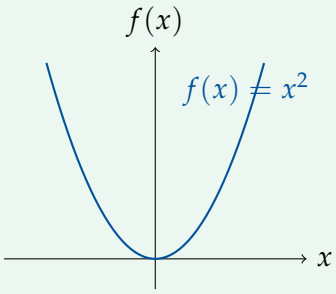
$$f(x) = f(-x)$$

Remark

Even functions are symmetric with respect to the y-axis.

Example 4.4: Even Function Example

$f(x) = x^2$



Definition 4.4: Odd Functions

A function $f : \mathbb{R} \to \mathbb{R}$ is odd if for all $x \in \mathbb{R}$,

$$f(-x) = -f(x)$$

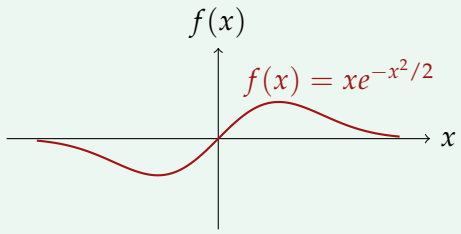
Remark

In odd functions, the positive part of the function is flipped in both the x and y axes (symmetric with respect to the origin).

Example 4.5: Odd Function Example

$f(x) = x \exp\left(-\frac{x^2}{2}\right)$
For $-x$, we have:

$$f(-x) = -x \exp\left(-\frac{(-x)^2}{2}\right) = -x \exp\left(-\frac{x^2}{2}\right) = -f(x)$$



Remark

- Integration of even functions give odd function (i.e $\int x^4 = \frac{1}{5}x^5$)
- Integration of odd functions give even function (i.e $\int x^3 = \frac{1}{4}x^4$)
- Definite integral of odd functions equal zero

Definite integral of odd function equal zero

Proof:

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$$

To evaluate $\int_{-a}^{0} f(x)\,dx$, let $u = -x$, then $dx = -du$,lower bound $u = a$ and upper bound $u = 0$:

$$\begin{aligned}\int_{-a}^{0} f(x)\,dx &= \int_{a}^{0} f(-u)\,(-du) \\ &= \int_{0}^{a} f(-u)\,du \\ &= -\int_{0}^{a} f(u)\,du\end{aligned}$$

We can replace u with x without changing the definite integrals value

$$\int_{-a}^{0} f(x)\,dx = -\int_{0}^{a} f(u)\,du = -\int_{0}^{a} f(x)\,dx$$

Finally, substitute this result back,

$$\int_{-a}^{a} f(x)\,dx = \left(-\int_{0}^{a} f(x)\,dx\right) + \int_{0}^{a} f(x)\,dx = 0$$

■

Remark

- For odd functions, integral for the negative part of function cancels the positive part.
- For even function the integral can be written $\int_{-a}^{a} f(x) = 2 * \int_{-a}^{0} f(x)\,dx = 2 * \int_{0}^{a} f(x)\,dx$

4.4 Fundamental Theorem of Calculus

Theorem 4.1: Fundamental Theorem of Calculus

For a continuous function $f : [a, b] \to \mathbb{R}$ and $x \in [a, b]$,

$$f(x) = \frac{d}{dx} \int_{a}^{x} f(t)\,dt$$

Remark

This theorem shows the relationship between derivatives and integrals. Simply stated, integrals and derivatives cancel each other out.

Example 4.6: Applying FTC

Let $f(x) = e^{-x}$ where $x > 0$. First, let's find the integral:

$$F(x) = \int_{0}^{x} e^{-t}\,dt = \left[-e^{-t}\right]_{0}^{x} = -e^{-x} - (-1) = 1 - e^{-x}$$

Taking the derivative of $F(x)$:

$$\frac{d}{dx}\left(1 - e^{-x}\right) = 0 - (-e^{-x}) = e^{-x} = f(x)$$

Corollary 4.1: Chain Rule for Integrals (Leibniz Rule)

If $f : [a, b] \to \mathbb{R}$ is a continuous function and $g : \mathbb{R} \to [a, b]$ is a differentiable function, then for all $x \in (a, b)$:

$$\frac{d}{dx} \int_{a}^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

Example 4.7: Differentiating the Gaussian CDF

Evaluate $\frac{d}{dx}\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{t^2}{2\sigma^2}\right)dt$.
Using the corollary where $g(x) = x$ and $g'(x) = 1$, the derivative evaluates directly to the inner function evaluated at x:

$$\frac{d}{dx}\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{t^2}{2\sigma^2}\right)dt = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{x^2}{2\sigma^2}\right)$$

Part II
Set Theory

Probability is defined as measure of sets. In probability we want to measure the size of sets and the theory for these sets is important. Since we are measuring sets, when calculating probability, often times we need to combine, split or remove sets. Operations are formalized in set theory.

5 Concepts of Sets

5.1 Set

Definition 5.1: Set

Set is an unordered collection of elements,

$$A = \{\xi_1, \xi_2, \ldots, \xi_n\}$$

These elements can be anything: numbers, objects, letters, functions etc.

Example 5.1: Set Examples

1. $\mathbb{R}$ is a set of real numbers.
2. $\mathbb{R}^2$ is a set of all real numbers ordered pairs.
3. $\{\text{apple, orange, banana}\}$ set of fruits.
4. L is a set of functions for straight lines that go through the origin. $\{f : \mathbb{R} \to \mathbb{R} \mid f(x) = ax\}$.

5.2 Subset

Definition 5.2: Subset

The set B is a subset of A, if any element $\xi \in B$ then ξ also belongs to A. Notation: $B \subseteq A$.
The equality of subsets are defined using subset, $A = B$, iff $B \subseteq A$ and $A \subseteq B$.

From that definition, subsets are divided into 2 categories:

1. **Proper subsets:** $B \neq A$
2. **Improper Subset:** $B = A$

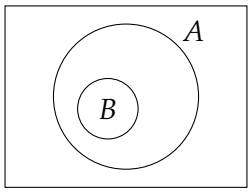


Figure 1: Visualization of proper and improper subsets

Definition 5.3: Empty Set

A set containing no elements is an empty set. Notation: $\emptyset$. The empty set is a subset of all sets: $\emptyset \subseteq A$, for any A. (Even if A is empty). $\{0\}$ is not an empty set.

Definition 5.4: Universal Set

A set containing all elements under consideration is a universal set. The definition of universal set is relative to the context of problem. For example: universal set can be $\{\text{Head, Tail}\}$ or $\{0, 1, 2, 3, 4, 5, 6\}$ or anything else. Notation: Ω.

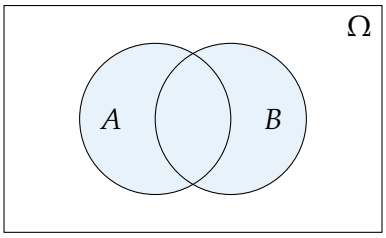
5.3 Set Operations

Definition 5.5: Union

The union of two sets is a new set containing all elements of the two.

$$A \cup B = \{\xi \mid \xi \in A \text{ or } \xi \in B\}$$

Venn diagram of union of two sets: The shaded region is $A \cup B$



Remark

Union of infinite sequences of sets

- Sequences are functions that maps integers to the set S. Example: $a_n = 2^n$ is $1, 2, 4, 8, \ldots$
- In the infinite sequence defined by $a_n = \frac{1}{2^n}$, we can evaluate a_∞ as 0. For evaluation using the $\lim_{n\to\infty} a_n = 0$ we can evaluate it because for finite and large n we see a_n converges to 0.
- We evaluate a_∞ by evaluating a_n at very large n.
- In the same way, the elements present in union of infinite sequence of sets must belong to one of A_n sets.

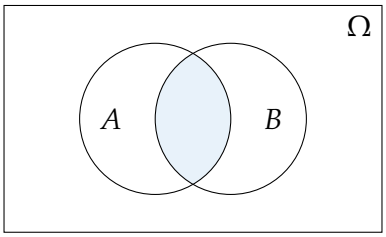
Mathematically, $\bigcup_{n=1}^{\infty} A_n$ for any finite large n.

Definition 5.6: Intersection

The intersection between two sets A and B, is the new set containing elements that belong to both A and B.

$$A \cap B = \{\xi \mid \xi \in A \text{ and } \xi \in B\}$$

The shaded region is $A \cap B$:



Remark

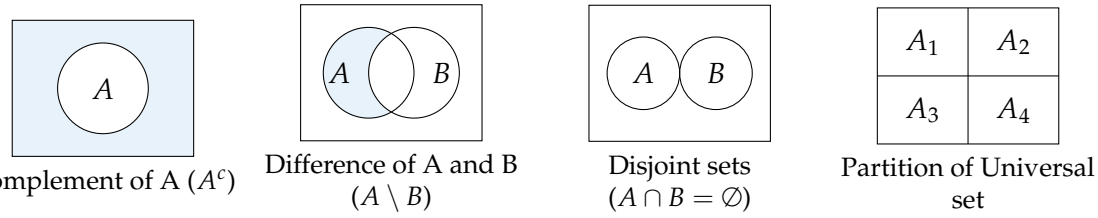
Intersection of infinite sequence of sets:
Using the same logic of union of infinite sequence of sets,

$$\bigcap_{n=1}^{\infty} A_n = \{\xi \mid \xi \in A_n \text{ for every finite } n\}$$

Definition 5.7: Complement

Complement of set A is all the elements of universal set except the ones present in A.

$$A^c = \{\xi \in \Omega \text{ and } \xi \notin A\}$$



Complement of A (A^c) | Difference of A and B ($A \setminus B$) | Disjoint sets ($A \cap B = \emptyset$) | Partition of Universal set

Figure 2: Visualization of complement of A, difference of A and B, disjoint sets and partition of universal set

Definition 5.8: Difference

The difference $A \setminus B$ is defined by,

$$A \setminus B = \{\xi \mid \xi \in A \text{ and } \xi \notin B\} = A \cap B^c$$

Definition 5.9: Disjoint

Two sets A and B are disjoint if,

$$A \cap B = \emptyset$$

Accordingly, a set of sets $A_1, A_2, \ldots, A_n$ are disjoint if,

$$A_i \cap A_j = \emptyset \text{ where } i \neq j$$

Definition 5.10: Partition

The collection of sets $\{A_1, A_2, \ldots, A_n\}$ form partition of universal set if,

1. $A_i \cap A_j = \emptyset$, for any pair $i \neq j$
2. $A_1 \cup A_2 \cup \cdots \cup A_n = \Omega$

5.4 Theorems of set operations

Theorem 5.1: Theorems concerning set operations

1. **Commutative:** $A \cap B = B \cap A$ and $A \cup B = B \cup A$
2. **Associative:** $A \cap (B \cap C) = (A \cap B) \cap C$ and $A \cup (B \cup C) = (A \cup B) \cup C$
3. **Distributive:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
4. **De Morgan's Law:** $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$

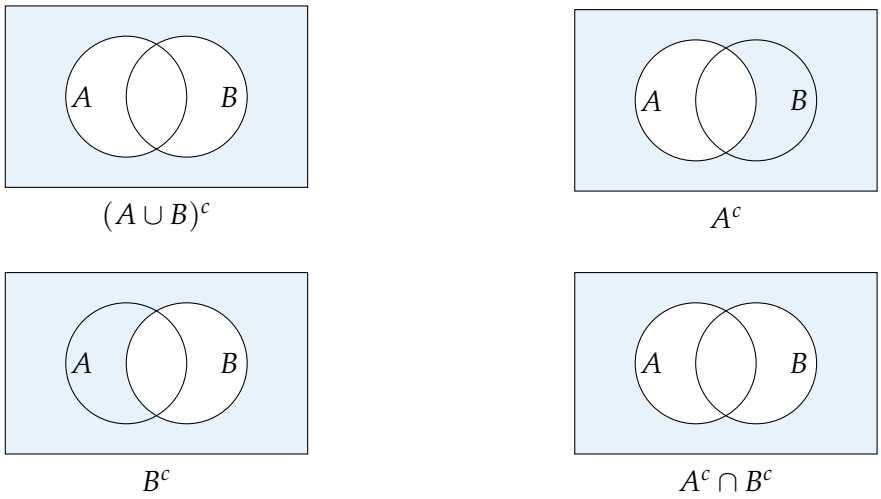


Figure 3: **Fig: Visualization of De Morgan's Law** $(A \cup B)^c = A^c \cap B^c$

Part III
Probability Basics

6 Probability Concepts

Definition 6.1: Probability

Probability is the measure of the size of a set. Size can mean count, interval (length), volume, or area. Probability of A is the belief A will occur in a random experiment.

Example 6.1: Probability of a Coin Toss

Probability of getting Heads (H) in a fair coin toss:

$$P(\{H\}) = \frac{|\{H\}|}{|\{T,H\}|} = \frac{1}{2}$$

Remark

- $P(A) = 0$ - Extremely minimal chances for A to occur but not impossible
- $P(A) = 1$ - Extremely high chances of A but not absolutely certain

6.1 Probability Space

Probability space definition is a precursor to defining probability for a specific event. It consists of the Sample Space, Event Space, and Probability Law.

Definition 6.2: Sample Space

Set of all possible outcomes of a random experiment. Notation: Ω.

Remark

- Sample space is collectively exaustive
- Sample space is mutually exclusive

Example 6.2: Dice Roll Sample Space

For a standard dice roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$

Definition 6.3: Event

A subset of the sample space. $E \subseteq \Omega$. Note that an event can be discrete, continuous, a function, etc.

Example 6.3: Even Dice Roll Event

Getting an even number on a dice roll: $E = \{2, 4, 6\} \subseteq \Omega$

Definition 6.4: Event Space

Set of all possible events (forms the event space). Event space is the power set of the sample space Ω. Notation: $\mathcal{F}$.

Example 6.4: Event Space for Coin Flip

For a coin flip where $\Omega = \{H, T\}$:

$$\mathcal{F} = \{\{H\}, \{T\}, \{H, T\}, \emptyset\}$$

Note: $H \neq \{H\}$ (one is an element, the other is a set).

Definition 6.5: Probability Law

Probability Law is a function that maps events to $[0, 1]$. $P : \mathcal{F} \to [0, 1]$.

Example 6.5: Probability Law Examples

- For an unbiased coin: $P(H) = \frac{1}{2}, P(T) = \frac{1}{2}$.
- For a uniform distribution: $P(E) = \frac{\text{Size of } E}{\text{Size of } \Omega}$.

Definition 6.6: Sigma Field

A collection of sets $\mathcal{F}$ becomes a Sigma field if:

1. If $F \in \mathcal{F}$ then $F^c \in \mathcal{F}$
2. If $F_1, F_2, \cdots \in \mathcal{F}$, then $\bigcup F_i \in \mathcal{F}$ and $\bigcap F_i \in \mathcal{F}$

Here $\mathcal{F}$ is a collection of sets, and F_i is a single set. A Sigma field implies "no holes"—all events are assigned probabilities.

Example 6.6: Sigma Field of Coin Toss

The Event space for $\{H, T\}$ is a Sigma field: $\{\{H\}, \{T\}, \{H, T\}, \emptyset\}$.

6.2 Discrete Uniform Law

Definition 6.7: A probability law

Discrete uniform law is a probability law P that assigns equal probability to each outcome. Probability of A containing n outcomes,

$$P(A) = \frac{\text{Number of outcomes in A}}{\text{Total number of outcomes } (|\Omega|)}$$

6.3 Axioms of Probability

Definition 6.8: Probability Axioms

Probability axioms are assumed to be true.

1. $P(E) \geq 0 \quad \forall E \in \mathcal{F}$
2. $P(\Omega) = 1$
3. $P(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$, where $A_1, A_2, \ldots, A_n$ are disjoint events.

6.4 Properties of Probability

Property 1: $P(A^c) = 1 - P(A)$ **Proof.** Here A and A^c are disjoint, and $A^c \cup A = \Omega$. Taking the probability of both sides:

$$P(A^c \cup A) = P(\Omega) = 1$$

Since they are disjoint, by Axiom 3:

$$P(A^c) + P(A) = 1 \implies P(A^c) = 1 - P(A)$$

■

Property 2: $P(A) \leq 1$ **Proof.** If A is an improper subset ($A = \Omega$), then $P(A) = 1$. If A is a proper subset, then A is contained in Ω, so $P(A) < 1$. Thus, $P(A)$ cannot be greater than 1. ■

Property 3: $P(\emptyset) = 0$ **Proof.** For two disjoint sets $\emptyset$ and Ω, $\emptyset \cup \Omega = \Omega$.

$$P(\emptyset \cup \Omega) = P(\emptyset) + P(\Omega)$$
$$P(\Omega) = P(\emptyset) + P(\Omega)$$

Since $P(\Omega) = 1$, we have $1 = P(\emptyset) + 1 \implies P(\emptyset) = 0$. ■

Additional Properties:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) \leq P(A) + P(B)$
- $P(A) \leq P(B)$ if B contains A ($A \subseteq B$)

Example 6.7: Probability Manipulation Examples

- $P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B)$
- $P(A \cap B^c) = P(A) - P(A \cap B)$

7 Conditional Probability

Definition 7.1: Conditional Probability

Given $A, B \in \mathcal{F}$ and $P(B) \neq 0$, the conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

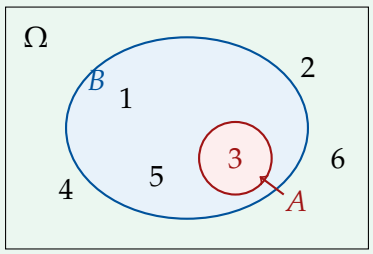
The chance element from A that has of happening in the new sample space B.

Example 7.1: Conditional Dice Roll

In a dice roll, say we have two events:

- $B = \{\text{Get odd numbers}\} = \{1, 3, 5\}$
- $A = \{\text{Get } 3\} = \{3\}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{3/6} = \frac{1}{3}$$



Restricting Sample Space to B

7.1 Conditional Probability and Axioms

We restrict the sample space in both LHS and RHS. The axioms still hold:

Axiom 1: $P(A|B) \geq 0$

Proof. $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Since $P(A \cap B) \geq 0$ and $P(B) > 0$, then $P(A|B) \geq 0$.

Axiom 2: $P(\Omega|B) = 1$

Proof. Probability of sample space is 1 even in restricted sample space:

$$P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

Axiom 3: $P(A \cup C|B) = P(A|B) + P(C|B)$ for disjoint A, C.

Proof.

$$P(A \cup C|B) = \frac{P((A \cup C) \cap B)}{P(B)} = \frac{P((A \cap B) \cup (C \cap B))}{P(B)}$$

Since A and C are disjoint, $A \cap B$ and $C \cap B$ are also disjoint. According to axiom 3:

$$\frac{P(A \cap B) + P(C \cap B)}{P(B)} = \frac{P(A \cap B)}{P(B)} + \frac{P(C \cap B)}{P(B)} = P(A|B) + P(C|B)$$

Probability of union of two disjoint sets in the restricted sample space is equal to the sum of probabilities of individual disjoint sets in the restricted sample space.

8 Independence

Definition 8.1: Independence of Events

Two events A and B are independent if and only if:

$$P(A \cap B) = P(A)P(B)$$

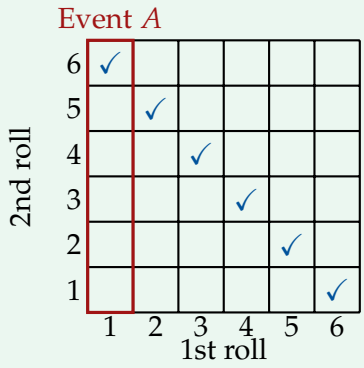
Example 8.1: Dice Rolled Twice

A dice is rolled twice.

- $A = \{\text{1st roll is 1}\}$
- $B = \{\text{sum is 7}\}$

$$P(A) = \frac{1}{6}$$
$$P(B) = \frac{6}{36} = \frac{1}{6}$$
$$P(A \cap B) = \frac{1}{36}$$

Since $\frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$, $P(A \cap B) = P(A)P(B)$. The events A and B are independent. Notice the event A has no influence on event B, but they are still independent.



Theorem 8.1: Conditional Probability of Independent Events

If A and B are independent events, then:

$$P(A|B) = P(A)$$

Proof.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

■

9 Bayes Theorem and law of total probability

Theorem 9.1: Bayes Theorem

If $P(A)$ and $P(B)$ are greater than zero,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Proof. We know $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$.

$$\implies P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

■

9.1 Law of total probability

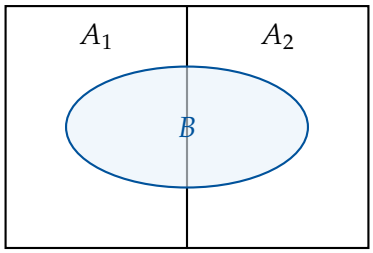
Definition 9.1: Law of Total Probability

If the following conditions satisfy for $\{A_1, A_2, \ldots, A_n\}$:

1. $A_i \cap A_j = \emptyset$ for any $i \neq j$ (Disjoint events)
2. $A_1 \cup A_2 \cup \cdots \cup A_n = \Omega$ (Events form partition)
3. $P(A_1) + P(A_2) + \cdots + P(A_n) = P(\Omega) = 1$

Then, we can write:

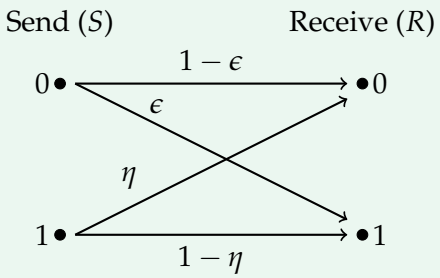
$$P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$



$P(B) = P(B \cap A_1) + P(B \cap A_2)$

Example 9.1: Binary Communication Channel

Over a channel, 0 and 1 are sent and received. Their probabilities are given in the figure:



Details of problem:

- $P[\text{Send } 1] = P(S = 1) = p$
- $P[\text{Receive } 1|\text{Send } 1] = P(R = 1|S = 1) = 1 - \eta$

Note: $(S = 1) \cup (S = 0) = \Omega$ and $(S = 1) \cap (S = 0) = \emptyset$.
Using the Law of Total Probability to find $P[\text{Receive } 1]$:

$$\begin{aligned} P[\text{Receive } 1] &= P(R = 1|S = 1)P(S = 1) + P(R = 1|S = 0)P(S = 0) \\ &= (1 - \eta)p + \epsilon(1 - p) \end{aligned}$$

Using Bayes Theorem to find $P[\text{Send } 1|\text{Receive } 1]$:

$$\begin{aligned} P[\text{Send } 1|\text{Receive } 1] &= \frac{P(R = 1|S = 1)P(S = 1)}{P(R = 1)} \\ &= \frac{(1 - \eta)p}{(1 - \eta)p + \epsilon(1 - p)} \end{aligned}$$

Part IV
Discrete Random Variables

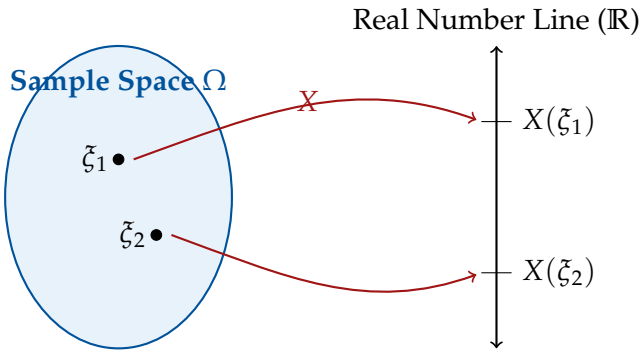
10 Random Variables

Definition 10.1: Random Variable

A random variable is a function that maps events in the sample space to the real number line ($\mathbb{R}$).

Remark

- Random variables help produce compact notations. For instance, $X(\xi) \in \mathbb{R}$.
- To use a random variable X, it needs to be defined. Meaning, it needs to be stated how events will be mapped to numbers.



10.1 Probability of Random Variable

Definition 10.2: Probability of a Random Variable

The probability of a random variable X taking a specific value a is the probability of the set of events associated with that value:

$$P(X = a) = P(\{\xi \mid \xi \in X^{-1}(a)\})$$

Example 10.1: Random Variable for Coin Flips

For 2 fair coin flips, the sample space is:

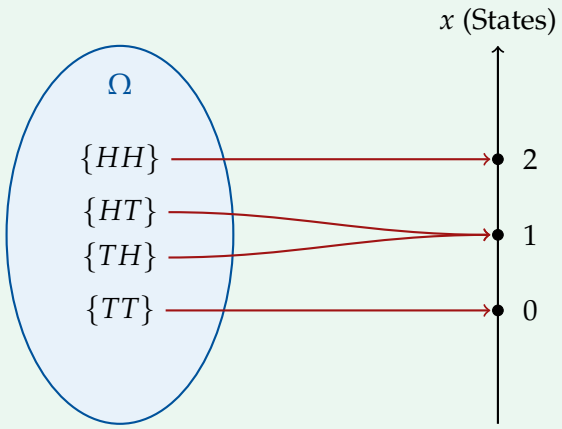
$$\Omega = \{HH, HT, TH, TT\}$$

Let the events be $E_1 = \{HH\}$, $E_2 = \{HT\}$, $E_3 = \{TH\}$, and $E_4 = \{TT\}$.
Let X be the random variable defining the **sum of the number of heads**. The mapping is as follows:

- $X(E_1) = X(\{HH\}) = 2$
- $X(E_2) = X(\{HT\}) = 1$
- $X(E_3) = X(\{TH\}) = 1$
- $X(E_4) = X(\{TT\}) = 0$

Thus, the random variable X has 3 states: 0, 1, and 2.
To evaluate $P(X = 1)$, meaning the probability of getting exactly one head from two flips:

$$P(X = 1) = P(\{\xi \mid \xi \in X^{-1}(1)\}) = P(\{HT, TH\}) = \frac{2}{4} = \frac{1}{2}$$



10.2 Probability Mass Function (PMF)

Definition 10.3: Probability Mass Function

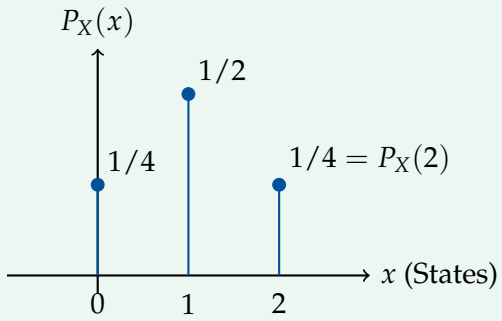
The probability mass function of a discrete random variable is the probability of obtaining state a.

$$P_X(a) = P[X = a]$$

Example 10.2: PMF of 2 Coin Flips

Continuing the previous example for X (number of heads in 2 fair coin flips), the PMF is:

$$P_X(0) = \frac{1}{4}$$
$$P_X(1) = \frac{1}{2}$$
$$P_X(2) = \frac{1}{4}$$



Remark

Ordinary vs. Random Variable:

- **Ordinary Variable:** Has one fixed deterministic value. (e.g., $2x + 4 = 0$). No PMF applies.
- **Random Variable:** Can take many values for different trials. Characterized by a PMF.

Histogram vs. PMF:

- **Histogram:** Generated empirically from a dataset. It is non-parametric. When the number of trials $n \to \infty$, the histogram converges to the PMF.
- **PMF:** Generated from the mathematical definition of the random variable. It is the parametric "ideal histogram" used to make statistical inferences.

Theorem 10.1: Property of PMF

The sum of probabilities of all states in the sample space is equal to 1.

$$\sum_{x} P_X(x) = 1$$

10.3 Cumulative Distribution Function (CDF)

Definition 10.4: Cumulative Distribution Function

The CDF of a random variable X is a function that maps states to real numbers, defined as the probability that X will take a value less than or equal to x_k:

$$F_X(x_k) = P[X \leq x_k] = P[X = x_1 \cup X = x_2 \cup \cdots \cup X = x_k] = \sum_{x' \leq x_k} P_X(x')$$

Remark

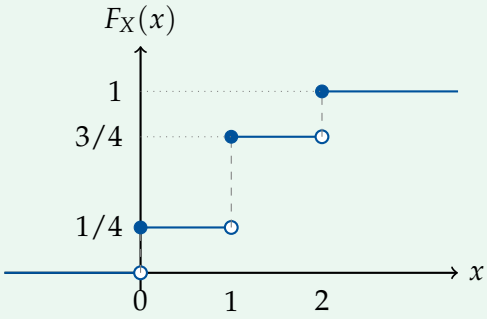
CDF for a specific state is found by summing the probabilities of all prior states and the probability of that state.
PMF vs CDF:

- The PMF is not a continuous function; it is rather a ”train of impulses” only applicable in discrete cases.
- The CDF is a well-defined function applicable in both discrete and continuous setups.

Example 10.3: CDF of 2 Coin Flips

Given $P_X(0) = \frac{1}{4}$, $P_X(1) = \frac{1}{2}$, and $P_X(2) = \frac{1}{4}$:

$$F_X(0) = \frac{1}{4}$$
$$F_X(1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$
$$F_X(2) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$



10.4 Properties of CDF

Theorem 10.2: Properties of CDF

1. The CDF is a sequence of non-decreasing steps.
2. $F_X(+\infty) = P[X \leq +\infty] = P[\Omega] = 1$. (All states less than or equal to infinity comprise the entire sample space).
3. $F_X(-\infty) = P[X \leq -\infty] = P[\emptyset] = 0$.
4. At x where $P_X(x) > 0$, the CDF has a vertical jump of height $P_X(x) = P[X = x]$. For discrete CDFs, the solid circle is on the jump (right-continuous) and hollow on the left.
5. **Converting CDF to PMF:**

$$P_X(x) = F_X(x) - F_X(x-1)$$

10.5 Expectation

Remark

Average from Histograms: For a dataset, the average is computed as:

$$Avg(H) = \sum_k x_k \frac{\text{frequency of } x_k}{\text{Total samples}}$$

Definition 10.5: Expectation

The expectation (or expected value) of a discrete random variable X is:

$$E[X] = \sum_{x \in X(\Omega)} x \cdot P_X(x)$$

Remark

The expectation formula is mathematically similar to the histogram average formula, but it is defined for the theoretical random variable X, not for an empirical dataset.

Example 10.4: Expected Value Example

Given $P_X(0) = \frac{1}{4}$, $P_X(1) = \frac{1}{2}$, and $P_X(2) = \frac{1}{4}$:

$$E[X] = 0\left(\frac{1}{4}\right) + 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{4}\right) = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

Example 10.5: Expected Salary

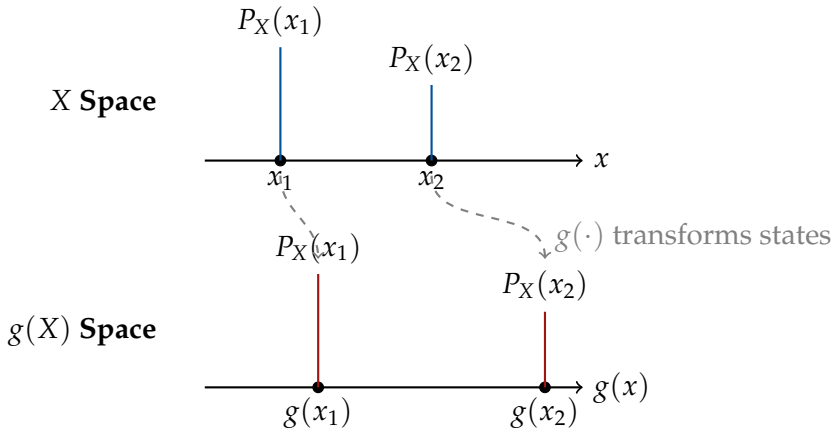
There are 3 employees with different salaries but the same role: \$10,000, \$20,000, and \$30,000. The probabilities of these salaries are $P_X(10000) = \frac{1}{2}$, $P_X(20000) = \frac{1}{4}$, and $P_X(30000) = \frac{1}{4}$.
If you join as a new employee, your expected salary is:

$$E[X] = 10000\left(\frac{1}{2}\right) + 20000\left(\frac{1}{4}\right) + 30000\left(\frac{1}{4}\right) = 5000 + 5000 + 7500 = 17,500$$

Remark

Expectation vs Sample Mean:

- **Expectation:** True mean of the random variable. It is a fixed, deterministic value. It might not exist for some specific distributions.
- **Sample Mean:** Mean calculated from a dataset ($\bar{x}$). The value changes as more data is added. As $n \to \infty$, the sample mean converges to the Expectation.



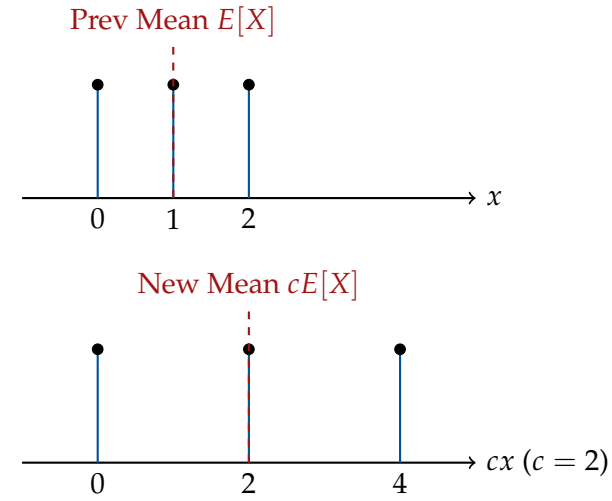
Probabilities stay unchanged, only states are transformed.

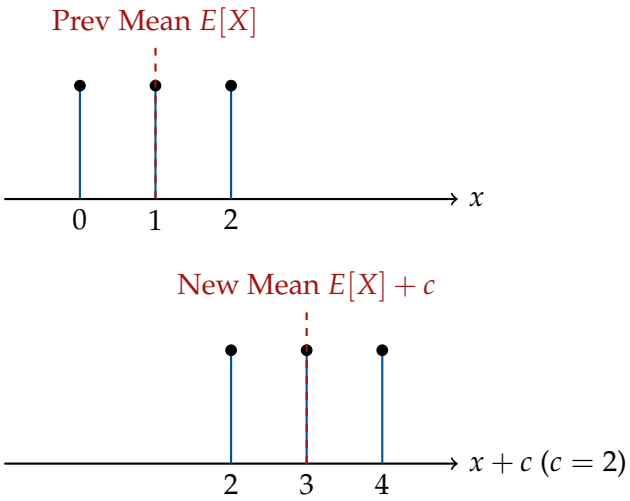
10.6 Properties of Expectation

Theorem 10.3: Properties of Expectation

- **Functions of a RV:**
$$E[g(X)] = \sum_x g(x)P_X(x)$$
The function transforms the state space, but $P_X(x)$ remains unchanged. The probability of obtaining the state does not change.
- **Linearity:** $E[g(X) + h(X)] = E[g(X)] + E[h(X)]$
- **Scaling:** $E[cX] = cE[X]$
- **Shift:** $E[X + c] = E[X] + c$

10.7 Moments and Variance





10.8 Moments

Definition 10.6: Moments

The k-th moment of a random variable X is $E[X^k]$.

Example 10.6: Second Moment of Coin Toss

Flip a coin 3 times. Let X be the number of heads. $P_X(0) = \frac{1}{8}$, $P_X(1) = \frac{3}{8}$, $P_X(2) = \frac{3}{8}$, $P_X(3) = \frac{1}{8}$.
The second moment ($k = 2$) is:

$$\begin{aligned} E[X^2] &= \sum x^2 P_X(x) \\ &= 0^2\left(\frac{1}{8}\right) + 1^2\left(\frac{3}{8}\right) + 2^2\left(\frac{3}{8}\right) + 3^2\left(\frac{1}{8}\right) \\ &= 0 + \frac{3}{8} + \frac{12}{8} + \frac{9}{8} = \frac{24}{8} = 3 \end{aligned}$$

10.9 Variance

Definition 10.7: Variance

The variance of a random variable X measures the spread of the data.

$$Var(X) = E[(X-\mu)^2] = \sum_x (x-\mu)^2 P_X(x)$$

Remark

- Squaring the deviation gets rid of the sign and heavily penalizes outliers.
- The state space of X is transformed to a squared deviation space, but the PMF stays unchanged.
- If states with higher deviations have a higher probability of happening, the RV has higher variance (data is more spread out).

10.10 Properties of Variancec

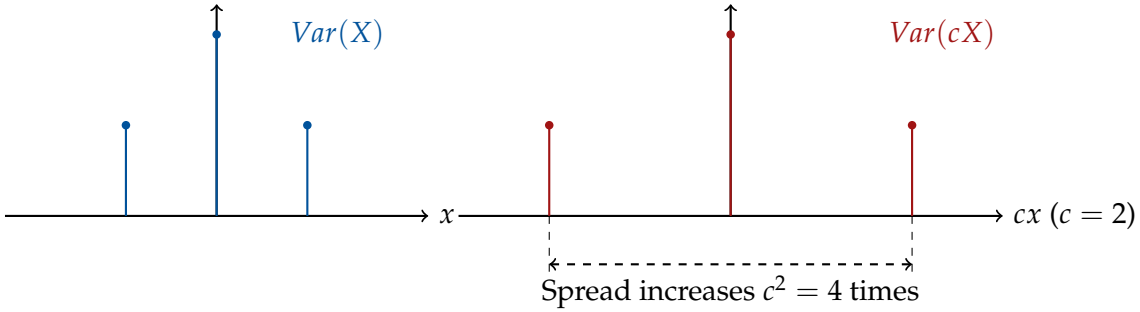
Theorem 10.4: Properties of Variance

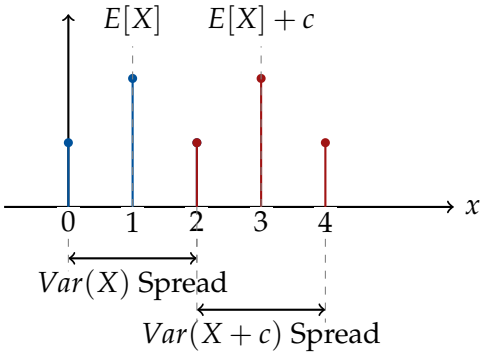
1. **Moment Formula:** $Var(X) = E[X^2] - (E[X])^2$
2. **Scaling:** $Var(cX) = c^2 Var(X)$
3. **Shift:** $Var(X + c) = Var(X)$

Proof of Moment Formula.

$$\begin{aligned}
Var(X) &= \sum_x (x^2 - 2\mu x + \mu^2) P_X(x) \\
&= \sum_x x^2 P_X(x) - 2\mu \sum_x x P_X(x) + \mu^2 \sum_x P_X(x) \\
&= E[X^2] - 2\mu(E[X]) + \mu^2(1) \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2 \\
&= E[X^2] - (E[X])^2
\end{aligned}$$

■





Spread stays the same ($Var(X + c) = Var(X)$)

Example 10.7: Variance of Bernoulli Trial

Let $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

$$E[X] = 1(p) + 0(1 - p) = p$$
$$E[X^2] = 1^2(p) + 0^2(1 - p) = p$$

Using the moment formula:

$$Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

10.11 Standard Deviation

Definition 10.8: Standard Deviation

The standard deviation of a random variable X is $\sqrt{Var(X)}$.

10.12 Standard Deviation

Definition 10.9: Linearity of Operators

An operator $L(\cdot)$ is linear if:

1. $L(u + v) = L(u) + L(v)$
2. $L(cu) = cL(u)$

The Expectation operator $E[\cdot]$ is a linear operator.

11 Bernoulli Random Variable

Definition 11.1: Bernoulli Random Variable

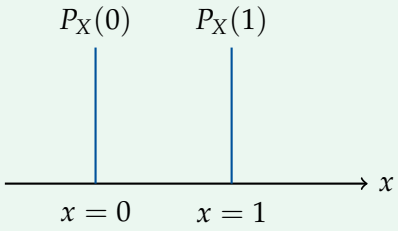
A Bernoulli random variable X,

$$P_X(0) = 1 - p \quad \text{and} \quad P_X(1) = p$$

where p is Bernoulli parameter and $0 < p < 1$.
X follows the Bernoulli distribution with probability parameter p. If an experiment is modeled such that there is success and failure and their probabilities are p and $(1 - p)$, then $X \sim \text{Bernoulli}(p)$.

Example 11.1: Fair coin toss

$X =$ Obtain head, $P_X(0) = \frac{1}{2}$, $P_X(1) = \frac{1}{2}$



$X \sim$ Bernoulli $\left(\frac{1}{2}\right)$

Example 11.2: Unfair coin toss

obtain head; $P_X(0) = \frac{1}{4}$, $P_X(1) = \frac{3}{4}$
we can say, $X \sim$ Bernoulli $\left(\frac{3}{4}\right)$
Behaves like known distribution.

11.1 Mean and Variance

Theorem 11.1: Mean

If $X \sim$ Bernoulli(p) then, $E[X] = p$.

Proof:

$$E[X] = \sum_x x P_X(x) = 0 \cdot P_X(0) + 1 \cdot P_X(1) = p$$

Theorem 11.2: Variance

If $X \sim$ Bernoulli(p) then, $Var(X) = p(1-p)$

Proof:

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] = E[(X-p)^2] \\ &= E[X^2 - 2Xp + p^2] \\ &= E[X^2] - 2pE[X] + p^2 \\ &= p - 2p^2 + p^2 \\ &= p - p^2 \\ &= p(1-p) \end{aligned}$$

Remark

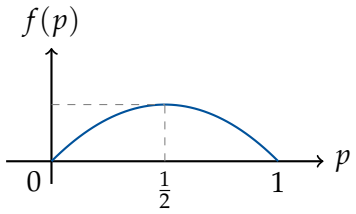
Maximum variance:

$$f(p) = p - p^2 \Rightarrow f'(p) = 1 - 2p$$

For critical point, $1 - 2p = 0 \Rightarrow p = \frac{1}{2}$
To check absolute max,

$$f''(p) = -2 \quad \text{which is } < 0.$$

So, max variance at $p = \frac{1}{2}$.



The change of probabilities mean change in expectation which means change in variance.

12 Binomial Random variable

Definition 12.1: Binomial Random variable

X is a Binomial random variable,

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$p \rightarrow$ Bernoulli parameter
$n \rightarrow$ Num. of trials.
Binomial RV counts the number of successes in n Bernoulli trials.
Binomial RV X can be written as sum of Bernoulli random variables,

$$X = I_1 + I_2 + \cdots + I_n$$

Example 12.1

Total num of success in n trials as X. $X =$ Number of heads in 3 coin flips, using definition,

$$P_X(2) = \binom{3}{2} p^2(1-p)^{3-2} = 3p^2(1-p)$$

Showing this probability calculation is correct.

$$P[\{HHT\} \cup \{HTH\} \cup \{THH\}] = P[\{HHT\}] + P[\{HTH\}] + P[\{THH\}]$$

Calculating for $P[\{HHT\}]$,

$$P[\{HHT\}] = P[\{H\}]P[\{H\}]P[\{T\}] \quad \text{(Independent states of Bernoulli RV)}$$

$$= P[I_1 = 1]P[I_2 = 1]P[I_3 = 0]$$
$$= p \cdot p \cdot (1-p) = p^2(1-p)$$

Similarly, $P[\{HHT\}] = P[\{HTH\}] = P[\{THH\}] = p^2(1-p)$
$P[\{HHT\} \cup \{HTH\} \cup \{THH\}] = 3p^2(1-p)$ Same as formula.

Remark

Components of the formula:

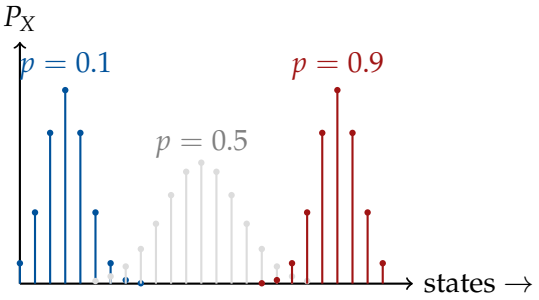
$$P_X(k) = \underbrace{\binom{n}{k}}_{\text{No of combinations}} \underbrace{p^k}_{\text{Probability of success}} \underbrace{(1-p)^{n-k}}_{\text{Probability of failure}}$$

12.1 Mean, Variance and Moment

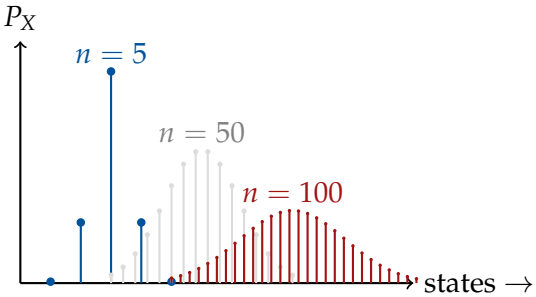
Theorem 12.1: Mean, Variance & moment

$X \sim \text{Binomial}(n, p)$
$n \to$ trials

1. $E[X] = E[I_1 + \cdots + I_n] = E[I_1] + \cdots + E[I_n] = np$
2. $E[X^2] = E[I_1^2 + \cdots + I_n^2] = E[I_1^2] + \cdots + E[I_n^2] = np$
3. $Var[X] = Var[I_1] + \cdots + Var[I_n] = np(1-p)$



(a) Varying p ($n = 60$).
The PMF shifts right because $E[X] = np$.
Variance increases then decreases (see graph $p(1-p)$).



(b) Varying n ($p = 0.5$).
For large n, it can be approx. by Gaussian.
Variance only increases. Density of states increases.

13 Geometric Random Variable

14 Poission Random Variable

Part V

Continuous Random Variables

Part VI

Joint Distributions

Part VII

Limit Theorems

Part VIII

Byesian Inference

Part IX

Classical Inference